

Machine Learning Internship Report

Intern Information

Name: Aashutosh Kumar

Domain: Machine Learning

Internship Organization: Aadyam Talent Consultancy

Internship Objective

The objective of this internship was to develop practical machine learning solutions by applying data preprocessing, visualization, feature engineering, model training, evaluation, and prediction techniques on real-world datasets.

Three industry-oriented machine learning projects were successfully completed covering Regression and Natural Language Processing.

Project 1 — House Rent Prediction

Objective

Develop a regression model capable of predicting house rent based on property-related features.

Dataset

- Total Records: 4746
- Features: 12
- Target Variable: Rent

Workflow

- Data Analysis
- Data Cleaning
- Feature Selection

- Label Encoding
- Linear Regression Model
- Prediction
- Visualization
- Model Saving

Technologies

- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib

Performance

- R^2 Score: **0.4401**
- MAE: **24408.85**

Deliverables

- Trained Model (.pkl)
 - Prediction Results
 - Visualization
 - Source Code
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Project 2 — Car Price Prediction

Objective

Predict used car selling prices using machine learning regression.

Dataset

- Original Records: 8128
- Processed Records: 6926

Workflow

- Duplicate Removal
- Missing Value Handling
- Label Encoding
- Random Forest Regression
- Prediction
- Visualization
- Model Saving

Technologies

- Python
- Pandas
- Scikit-learn
- Random Forest Regressor

Performance

- R^2 Score: **0.9195**
- MAE: **75316.21**
- RMSE: **132840.30**

Deliverables

- Prediction Results
- Trained Model
- Feature Importance Graph
- Actual vs Predicted Graph

Project 3 — Fake News Detection System

Objective

Develop a Natural Language Processing model capable of classifying news articles as Real or Fake.

Dataset

- Fake News: 23,478
- Real News: 21,211
- Total: 44,689

Workflow

- Dataset Merging
- Data Cleaning
- TF-IDF Vectorization
- Logistic Regression
- Prediction
- Classification Report
- Confusion Matrix
- Model Saving

Technologies

- Python
- Pandas
- Scikit-learn
- TF-IDF Vectorizer
- Logistic Regression

Performance

- Accuracy: **98.55%**
- Precision: **98.29%**
- Recall: **98.66%**
- F1 Score: **98.47%**

Deliverables

- Fake News Model
 - TF-IDF Vectorizer
 - Prediction Results
 - Classification Report
 - Confusion Matrix
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Skills Gained

- Data Cleaning
 - Data Preprocessing
 - Feature Engineering
 - Data Visualization
 - Regression
 - Classification
 - Natural Language Processing
 - Model Evaluation
 - GitHub Project Management
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Tools & Libraries

- Python
 - Pandas
 - NumPy
 - Scikit-learn
 - Matplotlib
 - Seaborn
 - Joblib
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Conclusion

This internship successfully demonstrated practical implementation of Machine Learning concepts through three real-world projects. The projects covered regression techniques for price prediction and natural language processing for fake news detection. The experience enhanced practical knowledge in data preprocessing, visualization, model development, evaluation, and deployment-ready project organization.

Intern: Aashutosh Kumar